



Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5	12 34 67 78 98
67 34 12 98 78	

Answer:

```

1  #include <stdio.h>
2
3  void swap(int *a, int *b) {
4      int temp = *a;
5      *a = *b;
6      *b = temp;
7  }
8
9  int partition(int arr[], int low, int high) {
10     int pivot = arr[high];
11     int i = low - 1;
12
13     for (int j = low; j < high; j++) {
14         if (arr[j] < pivot) {
15             i++;
16             swap(&arr[i], &arr[j]);
17         }
18     }
19     swap(&arr[i + 1], &arr[high]);
20     return i + 1;
21 }
22
23 void quickSort(int arr[], int low, int high) {
24     if (low < high) {
25         int pi = partition(arr, low, high);
26         quickSort(arr, low, pi - 1);
27         quickSort(arr, pi + 1, high);
28     }
29 }
30
31 int main() {
32     int n;
33     scanf("%d", &n);
34     int arr[n];
35     for (int i = 0; i < n; i++) {
36         scanf("%d", &arr[i]);
37     }
38     quickSort(arr, 0, n - 1);
39     for (int i = 0; i < n; i++) {
40         printf("%d ", arr[i]);
41     }
42     return 0;
43 }

```

Check

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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